PRISM INDUSTRIES PRIVATE LIMITED
AN ISO 9001:2015, ISO 14001:2015, ISO 45000:2018, WHO-GMP APPROVED COMPANY
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MATERIAL SAFETY DATA SHEET

(QUININE HYDROCHLORIDE)

SECTION 1 - CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product Name : Quinine Hydrochloride

Company Identification : Prism Industries Limited
Survey No.637/23/A1, Khambhat Dhuvaran Road, At: Kalamsar
Ta: Khambhat, Di: Anand-388640, Gujarat, India

Contact No. : 9727701668, 9712934397

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SECTION 2 - HAZARDS IDENTIFICATION

Classification of the substance or mixture
Classification according to Regulation (EC) No 1272/2008 (CLP)

Classification acc. to GHS			
Section	**Hazard class**	**Hazard class & Hazard category**	**Hazard Statement**
3.1O	acute toxicity (oral)	(Acute Tox. 4)	H302
3.4R	respiratory sensitisation	(Resp. Sens. 1A)	H334
3.4S	skin sensitisation	(Skin Sens. 1A)	H317

Section Hazard class Hazard class and category
Hazard

Remarks
For full text of Hazard- and EU Hazard-statements: see SECTION 16.
Label elements
Labelling according to Regulation (EC) No 1272/2008 (CLP)



Pictograms

Hazard statements

H302	Harmful if swallowed.
H317	May cause an allergic skin reaction.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.

Precautionary statements

Precautionary statements - prevention

P261	Avoid breathing dust/fume/gas/mist/vapours/spray.
P280	Wear protective gloves/protective clothing/eye protection/face protection.

Precautionary statements - response

P302+P352	IF ON SKIN: Wash with plenty of soap and water.
P333+P313	If skin irritation or rash occurs: Get medical advice/attention.
P342+P311	If experiencing respiratory symptoms: Call a POISON CENTER/doctor.

Precautionary statements - disposal

P501	Dispose of contents/container to industrial combustion plant.

Labelling of packages where the contents do not exceed 125 ml

Symbol(s)



H317	May cause an allergic skin reaction.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
P333+P313	If skin irritation or rash occurs: Get medical advice/attention.

Other hazards

There is no additional information.

SECTION 3 - COMPOSITION, INFORMATION ON INGREDIENTS

Substances

Name of substance	Quinine hydrochloride
EC number	231-437-7
CAS number	6119-47-7
Molecular formula	$C_{20}H_{25}ClN_2O_2$ * 2 H_2O
Molar mass	396,9 g/mol

SECTION 4 - FIRST AID MEASURES

Description of first aid measures



General notes
Take off contaminated clothing.
Following inhalation
Provide fresh air. In all cases of doubt, or when symptoms persist, seek medical advice.

Following skin contact
Rinse skin with water/shower. Brush off loose particles from skin. - In case of skin reactions, consult a
physician.

Following eye contact
Rinse cautiously with water for several minutes. In all cases of doubt, or when symptoms persist, seek
medical advice.

Following ingestion
Rinse mouth with water (only if the person is conscious). Call a doctor.

Most important symptoms and effects, both acute and delayed
After eye contact: Irritant effects, Conjunctival redness of the eyes,
Following skin contact: Localised redness, oedema, pruritis and/or pain, Allergic reactions,
After ingestion: Nausea, Vomiting, Gastrointestinal complaints,
Following inhalation: Cough

Indication of any immediate medical attention and special treatment needed
none

SECTION 5 - FIRE FIGHTING MEASURES

Extinguishing media
Suitable extinguishing media
Co-ordinate fire-fighting measures to the fire surroundings
water spray, foam, dry extinguishing powder, carbon dioxide (CO2)

Unsuitable extinguishing media
water jet

Special hazards arising from the substance or mixture
Combustible.

Hazardous combustion products
In case of fire may be liberated: nitrogen oxides (NOx), carbon monoxide (CO), carbon dioxide (CO2), hydrogen chloride (HCl)

Advice for firefighters
Fight fire with normal precautions from a reasonable distance. Wear self-contained breathing apparatus.

Special protective equipment for firefighters
Protective clothing against liquid and gaseous chemicals, including liquid aerosols and solid particles. Self-contained breathing apparatus (SCBA). Self-contained breathing apparatus (EN 133).

SECTION 6 - ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures
For non-emergency personnel
Wearing of suitable protective equipment (including personal protective equipment referred to under Section 8 of the safety data sheet) to prevent any contamination of skin, eyes and personal clothing. Do not breathe dust. Avoid contact with skin, eyes and clothes.

Environmental precautions
Keep away from drains, surface and ground water.

Methods and material for containment and cleaning up
Advices on how to contain a spill

Covering of drains.

Advices on how to clean up a spill

Take up mechanically. Control of dust.

Other information relating to spills and releases

Place in appropriate containers for disposal.

Reference to other sections

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

SECTION 7 - HANDLING and STORAGE

Precautions for safe handling

Provision of sufficient ventilation.

Measures to prevent fire as well as aerosol and dust generation

Removal of dust deposits.

Advice on general occupational hygiene

Wash hands before breaks and after work. Keep away from food, drink and animal feedingstuffs.

Conditions for safe storage, including any incompatibilities

Store in a dry place. May cause decomposition by long-term light influence. Keep container tightly closed.

Incompatible substances or mixtures

Observe hints for combined storage.

Consideration of other advice

Ventilation requirements

Use local and general ventilation.

Specific designs for storage rooms or vessels

Recommended storage temperature: 15 - 25 °C.

Specific end use(s)

No information available.

SECTION 8 - EXPOSURE CONTROLS, PERSONAL PROTECTION

Control parameters
National limit values
Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	Case No	Notation	Identifier	TWA [mg/m^3]	STEL [mg/m^3]	Source
JP	particulates not otherwise classified		dust	OEL	8		JSOH
JP	particulates not otherwise classified		r	OEL	2		JSOH
JP	dust		less3silica , dust	OEL	4		JSOH
JP	dust		less3silica , r	OEL	1		JSOH

Notation

dust As dust less3silica Containing less than 3% cry stalline silicar Respirable fraction
STEL Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period unless otherwise specified TWA Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours time-weighted average

Exposure controls
Individual protection measures (personal protective equipment)
Eye/face protection
Use safety goggle with side protection.
Skin protection
hand protection
Wear suitable gloves. Chemical protection gloves are suitable, which are tested according to EN 374.
For special purposes, it is recommended to check the resistance to chemicals of the protective gloves
mentioned above together with the supplier of these gloves.

type of material
NBR (Nitrile rubber)
material thickness
>0,11 mm.
breakthrough times of the glove material
>480 minutes (permeation: level 6)
other protection measures
Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

Respiratory protection
Respiratory protection necessary at: Dust formation. Particulate filter device (EN 143). P2 (filters at least 94 % of airborne particles, colour code: White.
Observe the wear time limits according GefStoffV in combination with the rules for using respiratory protection apparatus (BGR 190).
Environmental exposure controls
Keep away from drains, surface and ground water.

SECTION 9 - PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties
Appearance

Physical state	Solid (powder, crystalline)
Colour	white
Odour	odourless
Odour threshold	No data available

Other physical and chemical parameters

pH (value)	6 - 6,8 (water: 10 g/l, 20 °C)
Melting point/freezing point	not determined
Initial boiling point and boiling range	This information is not available.
Flash point	122 °C
Evaporation rate	no data available
Flammability (solid, gas)	Non-flammable
Explosive limits	
• lower explosion limit (LEL)	(30 g/m³)
• upper explosion limit (UEL)	this information is not available

Explosion limits of dust clouds	these information are not available
lower explosion limit (LEL)	30 g/m³
Vapour pressure	This information is not available.
Density	This information is not available.
Vapour density	This information is not available.
Bulk density	~ 310 kg/m³
Relative density	Information on this property is not available.
Solubility(ies)	
Water solubility	62,5 g/l at 25 °C
Partition coefficient	
n-octanol/water (log KOW)	1,78 (calculated value)
Auto-ignition temperature	Information on this property is not available.
Decomposition temperature	115 °C at 1.013 hPa
Viscosity	not relevant (solid matter)
Explosive properties	none
Oxidising properties	none

Other information

There is no additional information.

SECTION 10 - STABILITY AND REACTIVITY

Reactivity

The product in the delivered form is not dust explosion capable; the enrichment of fine dust however leads to the danger of dust explosion.

Chemical stability

The material is stable under normal ambient and anticipated storage and handling conditions of temperature and pressure.

Possibility of hazardous reactions

Violent reaction with: Ammonia (NH3), Iodine, Strong oxidiser, Strong alkali

Conditions to avoid

UV-radiation/sunlight. Humidity. Decompostion takes place from temperatures above: 115 °C at 1.013 hPa.

Incompatible materials

There is no additional information.

Hazardous decomposition products

Hazardous combustion products: see section 5.

SECTION 11 - TOXICOLOGICAL INFORMATION

Information on toxicological effects

Acute toxicity

Harmful if swallowed

Exposure route	Endpoint	Value	Species	Source
oral	LD50	620 mg/kg	rat	---

Skin corrosion/irritation

Shall not be classified as corrosive/irritant to skin.

Serious eye damage/eye irritation

Shall not be classified as seriously damaging to the eye or eye irritant.

Respiratory or skin sensitisation

May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause an allergic skin
reaction. May cause sensitisation by skin contact. May cause sensitization by inhalation.

Summary of evaluation of the CMR properties

Shall not be classified as germ cell mutagenic, carcinogenic nor as a reproductive toxicant

Specific target organ toxicity - single exposure

Shall not be classified as a specific target organ toxicant (single exposure).

Specific target organ toxicity - repeated exposure

Shall not be classified as a specific target organ toxicant (repeated exposure).

Aspiration hazard

Shall not be classified as presenting an aspiration hazard.

Symptoms related to the physical, chemical and toxicological characteristics

If swallowed

nausea, vomiting

If in eyes

slightly irritant

If inhaled

cough

If on skin

Prolonged or repeated contact with skin or mucous membrane result in irritation symptoms such as redness, blistering, dermatitis, etc

Other information

None

SECTION 12 - ECOLOGICAL INFORMATION

Toxicity
acc. to 1272/2008/EC: Shall not be classified as hazardous to the aquatic environment.

Process of degradability
Theoretical Oxygen Demand with nitrification: 2,066 mg/mg
Theoretical Oxygen Demand: 1,894 mg/mg
Theoretical Carbon Dioxide: 2,218 mg/mg

Bioaccumulative potential
Does not significantly accumulate in organisms.
n-octanol/water (log KOW) 1,78

Mobility in soil
Data are not available.

Results of PBT and vPvB assessment
Data are not available.

Other adverse effects
Slightly hazardous to water.

SECTION 13 - DISPOSAL CONSIDERATIONS

Waste treatment methods
This material and its container must be disposed of as hazardous waste. Dispose of contents/ container in accordance with local/regional/national/international regulations.

Sewage disposal-relevant information
Do not empty into drains.

Relevant provisions relating to waste
The allocation of waste identity numbers/waste descriptions must be carried out according to the EEC, specific to the industry and process.

Remarks
Waste shall be separated into the categories that can be handled separately by the local or national waste management facilities. Please consider the relevant national or regional provisions.

SECTION 14 - TRANSPORT INFORMATION

UN number	(not subject to transport regulations)
UN proper shipping name	not relevant
Transport hazard class(es)	not relevant
Class	-
Packing group	not relevant
Environmental hazards	none (non-environmentally hazardous acc. to the dangerous goods regulations)

Not regulated as per IATA

Special precautions for user

There is no additional information.

Transport in bulk according to Annex II of MARPOL and the IBC Code

The cargo is not intended to be carried in bulk.

Information for each of the UN Model Regulations

Transport of dangerous goods by road, rail and inland waterway (ADR/RID/ADN)

Not subject to ADR, RID and ADN.

International Maritime Dangerous Goods Code (IMDG)

Not subject to IMDG.

SECTION 15 - REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Relevant provisions of the European Union (EU)

Regulation 649/2012/EU concerning the export and import of hazardous chemicals (PIC)

Not listed.

Regulation 1005/2009/EC on substances that deplete the ozone layer (ODS)

Not listed.

Regulation 850/2004/EC on persistent organic pollutants (POP)

Not listed.

Restrictions according to REACH, Annex XVII

not listed

List of substances subject to authorisation (REACH, Annex XIV)

not listed

Directive 2011/65/EU on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS) - Annex II

not listed

Regulation 166/2006/EC concerning the establishment of a European Pollutant Release and Transfer Register (PRTR)

not listed

Directive 2000/60/EC establishing a framework for Community action in the field of water policy

(WFD)

not listed

National inventories

Substance is listed in the following national inventories:

EINECS/ELINCS/NLP (Europe)

Chemical Safety Assessment

No Chemical Safety Assessment has been carried out for this substance.

SECTION 16 - ADDITIONAL INFORMATION

The information above is believed to be accurate and represents the best information currently available to us. However, we make no warranty of merchantability or any other warranty, express or implied, with respect to such information, and we assume no liability resulting from its use. Users should make their own investigations to determine the suitability of the information for their particular purposes. In no way shall the company be liable for any claims, losses, or damages of any third party or for lost profits or any special, indirect, incidental, consequential or exemplary damages, howsoever arising, even if the company has been advised of the possibility of such damages